ASSIGNMENT – 3

Subject: PPS

One-Dimensional and Two-Dimensional Arrays in C [Submission date: 27th Oct, 2026]

Instruction: Write complete C programs. Read all required array elements and dimensions from the keyboard and display suitable output messages.

1. Read n integers into an array and find the largest and smallest elements.

2. Calculate the sum and average of all elements of an integer array.

3. Count how many odd and even numbers are present in an array.

4. A shop stores product identification numbers in an integer array. Search for a product ID entered by the user and display its position if found.

5. The marks of n students are stored in an array. Arrange the marks in ascending order.

6. Find the second-largest element of an array.

7. Find the frequency of a value entered by the user in an integer array.

8. Display the elements of an array in reverse order without using another array.

9. Read two matrices of the same order and display their sum.

10. Read a matrix and display its transpose.

11. Calculate and display the sum of elements of each row of a matrix.

12. Calculate and display the sum of elements of each column of a matrix.

13. Display the principal diagonal elements of a square matrix and calculate their sum.

14. Read two matrices of sizes m x n and p x q. Check whether their data can be combined through row-column processing. If possible, calculate and display the resultant matrix.

15. A retail company has several stores selling different products. The quantities sold are stored in one 2D array and corresponding price information in another. Write a program to calculate and display the resulting revenue information for the stores.

16. Find the largest element in a matrix and display its row and column position.

17. Check whether a given square matrix is symmetric.

18. Determine whether a given matrix is a sparse matrix.

19. Write a C program that reads an integer n and stores the first n Fibonacci numbers in an array.

$F_n = F_{n-1} + F_{n-2}$, $F_0 = 0$, $F_1 = 1$

20. Write a C program that reads an integer n and uses an array to efficiently find out the first n prime numbers.

21. Read in an integer n, read in n integers and print the integer with the highest frequency.

22. Read in an integer n, read in n numbers and find out the mean, median and mode.

23. Read in two integers, m and n. Read m integers in array X and n integers in B. Find out the common elements in arrays X & Y and store it those another array Z.

24. Read in two integers, m and n. Read m integers in array X and n integers in B. Find out the elements that are either in X or Y and store those in another array Z.

25. The digits of a mobile number are stored in an integer array. Write a C program to determine the digit occurring the maximum number of times.

Input: 7070254486

Output: Digit 0 occurs 3 times

Submit complete programs with proper logic, formatting, and sample output.